Paper 4

The Photon as Baseline, Electron-Positron as First Tilt

Paper 4 of the Series

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ABSTRACT

Paper 1 promised a universal ruler for mass, gravity, and time, and Paper 3 supplied the mechanism that ruler would measure — the axis spin that turns the photon into the electron and the positron. This paper joins the two. It establishes the photon as the ruler's zero and the electron-positron pair as its first resolved mark: the single fixed origin and the first self-contained departure against which every later quantity in the series — stability, motion, mass, gravity, time — is read. The photon is the zero because its inside carries nothing: pure O at the central junction, no departure from baseline, no mass, and no duration (it runs only the inner identity scan, never the temporal scan whose accumulation defines duration). The electron-positron pair is the first resolved mark because, although it is reached only after three axis tilts and not by a single tilt, it is the first balanced configuration the spin produces besides the photon — the first departure that holds rather than reaching for resolution. A single tilt off baseline produces an open state (State 7 or State 8); it takes three tilts to arrive at the first balanced state, and the paper proves this as a counting theorem: three tilts is the minimum number of center moves that can close a triplet lock while conserving the value pool, so any first resolved configuration must sit at least three steps from the photon. Together the photon and this first resolved pair fix the origin and the scale of the ruler, and the direction of its two signs. This paper sets that anchor and points forward; it does not take up the stability, mass-accounting, or time treatments assigned to later papers.

1. Purpose: The Necessity of an Absolute Zero

Paper 3 answered *how* a configuration departs from baseline: the axis spin relocates values between the inside and the outside, turning the photon into the electron and the positron. That was a mechanism. Paper 1 promised something the mechanism alone does not give — a *ruler*: a way to measure mass, gravity, and time against a common, non-arbitrary reference.

A ruler needs two things before it can measure anything: an absolute zero, and a first mark. The zero must be unalterable, or every measurement built on it inherits the wobble of a moving origin. This is the deepest requirement of an objective physics: a single origin from which all properties are read without the observer adjusting the origin to suit the measurement. A system without an absolute zero is a ledger whose ruler changes length depending on what is being weighed.

This paper supplies both fixed points. The zero is the photon. The first resolved mark is the electron-positron pair. Everything the later papers measure is read from this origin and scaled to this first departure. The purpose of Paper 4 is to fix the origin and the scale so the rest of the series has a stable thing to measure against — and to show that one medium with one baseline removes the need for the shifting, scale-dependent standards that a framework without an absolute zero is forced to invent. (The full critique of those shifting standards is the work of the companion paper, Paper 4.5; this paper states only the positive case.)

The contract of this paper. To make the division of labor explicit rather than implied: this paper *proves* that the photon is the unique non-arbitrary zero of the ruler and that State 1 is the unique first resolved mark, including the counting theorem of Section 3 (three tilts is the minimum that can close a triplet lock while conserving the value pool). It *assumes* from Papers 1–3 the three-axis framework, the Per-Triangle Rule, the nine-state classification, the inside-outside structure, the spin operation, and the two scans. It does *not* derive numerical masses, charges, or spin values — those calculations arrive in Papers 8 and 10, built on the counting rules fixed here — and it does not treat stability (Paper 5), time's full ledger (Paper 13), or the composite spectrum (Paper 16). What it provides is the anchor those papers require: an origin, a first mark, and the counting rule that makes the later derivations possible.

Figure 1. The Universal Ruler: Sovereign Zero and the First Resolved Marks

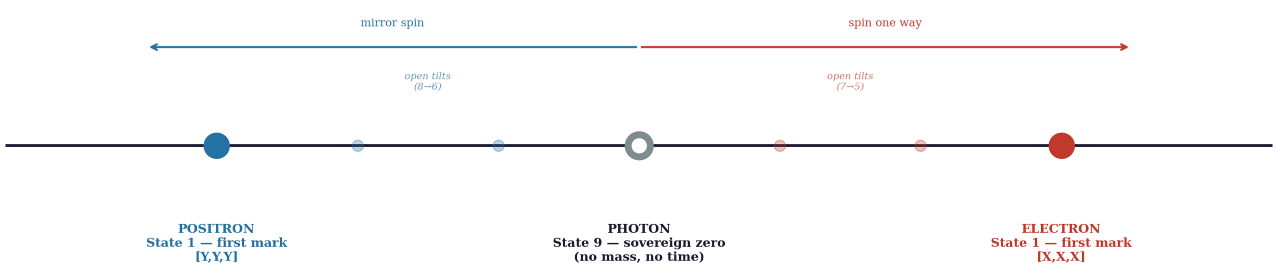

Figure 1. The universal ruler: the photon (State 9) as sovereign zero, the electron and positron (State 1) as the first resolved marks reached by the three-tilt spin in opposite rotational directions, with the open intermediate tilts shown as faint waypoints.

2. The Geometry of the Scan: Why the Photon Is the Zero of Mass *and* Time

Figure 2. The Photon Runs the Identity Scan Alone; A Resolved Mark Runs the Full 720° Manifold Scan

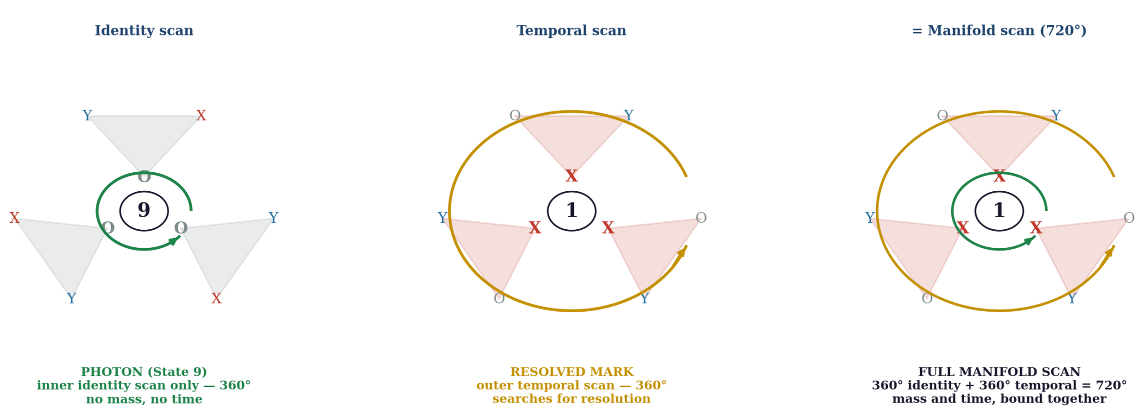

Figure 2. The photon runs the inner identity scan alone (360°, no mass, no time); a resolved mark also runs the outer temporal scan (360°); the two together form the full 720° manifold scan that binds mass and duration.

The photon is State 9: pure O at all three center positions, with three X and three Y balanced across its outer surface (Papers 2, 3). It carries no mass and travels at c. These facts are established. What this paper adds is *why that makes it the zero of the ruler, and why the zero could be nothing else* — and the answer runs through the two scans.

A baseline cannot be an arbitrary choice, or the ruler it founds is arbitrary too. The photon is the non-arbitrary zero because it is the one state with nothing registered on the inside. Its central junction is pure O — the absence of any departure. Every other state carries at least one active value at its center; every other state has already departed. The photon is the unique state from which departure has not begun, which is exactly what a zero must be: not a small quantity, but the absence of the quantity.

2.1 The Photon: Identity Scan Only, and Therefore Timeless

A configuration maintains itself by scanning. The inner **identity scan** is the self-contained loop by which a configuration reads its three center values against its six outer values and confirms what it is (Paper 3, Section 7A). The outer **temporal scan** is the loop by which a configuration carrying tilt searches the surrounding fabric for resolution, and the accumulation of those outer sweeps is what registers duration (Paper 3, Section 7B). The two together compose the full 720-degree manifold scan.

The photon's pure-O junction limits it to the inner identity scan alone. With no active value tilted into its center, it has no axis displacement to process and nothing unresolved to search the surroundings for, so it never runs the outer temporal scan. It performs only localized, self-contained identity checks. There is no sequential progression to accumulate; the photon exists in a permanent geometric *now*. This is why the photon is the zero of *time* as fully as it is the zero of mass — where "no time" means precisely: runs no temporal scan, therefore accrues no duration. Zero departure, zero mass, zero time — one structural fact, read three ways.

A definitional point should be fixed here, because it decides whether the claim is circular. Within this framework, time is not assumed as an independent background variable that things move through. Duration is *operationally defined* as the accumulation of completed temporal scans — that is the framework's definition of the quantity, stated once and held throughout the series. The statement "the photon accrues no duration" is therefore not a conclusion smuggled out of a definition; it is the definition applied: a configuration that runs no temporal scans accumulates zero of the quantity duration measures. Whether this operational definition is the right account of physical time is exactly what the series tests against observation; what it is not is circular.

The obvious objection is answered directly: the photon *moves*, and motion would seem to imply sequence and therefore time. It does not, because of what the motion is. Propagation at c is not a sequential displacement driven by the photon's own temporal processing; it is the passive, synchronous manifestation of the medium's native identity-scan rate acting upon its own baseline fabric — where "passive" means precisely: no energy is drawn from the photon's own ledger; the motion is the medium's clock, not the photon's labor. The photon does not work its way through the medium step by registered step — it *is* baseline fabric, gliding in sync with the identity scans the surrounding triangles are already running, and c is simply that universal processing rate made visible. A configuration registers time only by running its own temporal scan against the fabric; the photon, in sync with the fabric, has nothing to register and nothing to register it with.

A consequence of running no temporal scan, stated here as footwork and developed in the companion critique (Paper 4.5), concerns what a photon carries in flight. The claim must be made precisely, because made loosely it collides with both standard physics and this series' own mechanics. The photon runs only the identity scan, and therefore carries no proper-time clock of its own; it has no property that *the photon* *itself registers* in flight, because registration is the temporal scan's work and the photon runs none. This is not a denial that interactions involving photons exhibit the familiar quantities — energy, momentum, polarization; it is a relocation of where those quantities are registered. Any reading is made by a *target's* scan at an interaction endpoint: the outer geometry of other configurations evaluates a passing photon during their own scans (as the grazing case of Paper 4.5 describes), but the photon does not participate — unless a direct hit drives creation geometry, it passes unchanged. All familiar photon properties appear at interaction endpoints, registered by the interacting configuration's machinery, never read off the photon mid-flight by the photon itself. The standard relation between a photon's energy and what an apparatus measures is not contradicted by this; it is re-housed at the endpoint where the measurement actually occurs. The demonstration that the standard attribution of in-flight properties is an inference from the endpoint rather than an observation of the photon is the work of Paper 4.5.

2.2 The Resolved Mark: Full Manifold Scan, and Therefore Duration and Mass

The moment active values are tilted into the center positions, the configuration executes a definite departure from the photon baseline. In this framework mass is not an independent substance or auxiliary cargo brought into the system; all mass is a departure from the photon. The invariant energy of the baseline state is conserved through the transition — nothing is lost. When the configuration undergoes its axis tilts, that same conserved energy is redirected into the geometry of a resolved departure, where it registers as mass.

Because energy is conserved, the displacement triggers a mechanical unification within the three-axis hardware. The redirected energy drives the configuration to expand its operational loop into the full manifold scan. The manifold scan is not two separate systems running in isolation; it is the structural integration of the internal and external checks required to hold a state away from baseline — a complete 720-degree processing loop composed of two 360-degree phases. The internal phase is the identity scan: the inner 360-degree loop by which the configuration verifies its core center values against its surface geometry. The external phase is the temporal scan: the outer 360-degree loop, present only for resolution, by which the configuration audits the surrounding triangles of the continuous medium to reconcile its localized axis displacement. Together the inner identity scan and the outer temporal scan form the complete 720-degree manifold scan.

This operational link is the reason mass and time are bound together. Once the tilt occurs, the conserved energy must route through both halves of the 720-degree cycle to reach resolution, so what macroscopic physics calls "mass" and "duration" are two viewports onto one conserved geometric process. Mass is the ledger entry tracking how much photon energy has been transferred into the localized geometric displacement of the center positions — how heavily the medium is loaded. Duration is the sequential rate at which that same energy drives the temporal phase of the scan to search for resolution — how long the configuration has operated away from baseline. A configuration is therefore in exactly one of two conditions: running its inner identity loop alone, as a timeless, massless sovereign zero, or driving the full 720-degree manifold scan as a resolved departure that carries both mass and duration. There is no third option, and there is no mass without time or time without mass, because both are the same conserved currency read through two windows.

For precision, the currency itself should be named in this framework's own terms: energy is added geometry — the structural content delivered into the medium by an interaction, conserved absolutely as it relocates (the definition the series carries from its foundations). This closes a question the standard reader will otherwise ask — what drives the departure if mass is only its result — and it yields an explanation the framework gets for nothing. Once a configuration departs from the photon, the energy of the tilt is divided by duty: part of it is consumed by the temporal scan and the full manifold accounting that the departure now obligates. This is exactly why the speed drop off baseline is so severe — why the very first step away from the photon falls so far below c. The deficit is not lost to friction or radiated away; it is the portion of the conserved budget now spent running the temporal phase of the scan rather than driving propagation. The configuration moves slower because it is, in the most literal mechanical sense, busier: its energy is tied into scan accounting and into the mass of the loaded center, not into motion. Speed, mass, and duration are three withdrawals from one conserved account, and the photon alone spends everything on propagation because it has no scan debt to service.

Calculation deferred: the formal numbers are the work of later papers — Paper 8 calculates mass as the internal accounting of departure, and Paper 13 calculates time as the 720-degree rotation ledger, each from this same conserved currency and without a parameter tuned by hand. This paper fixes the structure those calculations run on.

3. The Minimal Resolved Departure: Electron-Positron as the First Mark

If the photon is the zero, the first mark is the smallest *resolved* departure from it — and here the precise correction of this paper matters most. The first mark is **not** the first tilt. A single axis tilt off the photon drives one center value from O to an active value and produces an **open** state — State 7 or State 8, depending on the axis — whose outer surface is lopsided and which therefore reaches for resolution rather than standing alone. An open state cannot be a unit; it is unsettled by definition.

The electron (inside X, X, X) and the positron (inside Y, Y, Y) — State 1 — are reached only after **three** tilts, through the conserving trajectory Paper 3 traced: photon to State 7 to State 5 to State 1, each step a real intermediate state, conservation of three X, three Y, three O holding throughout (Paper 3, Appendix C; re-grounded in Appendix A here). What makes State 1 the first mark is not that it is reached first in the count of tilts — it is reached *third* — but that it is the **first balanced, resolved configuration besides the photon**. It is the first departure that, once made, holds: a self-contained state with nothing unsettled on its surface, rather than an open state still searching.

This is what qualifies it as the ruler's first mark, and the criterion itself deserves justification rather than assertion — *why* should "resolved" be the test, rather than simply "first departure"? The answer is metrology, not preference. A ruler mark must be repeatable: the same place every time it is consulted, independent of circumstance. Open states are transitional conditions whose continued existence depends on unresolved interactions — what they are at any moment is entangled with what the surrounding fabric has and has not yet offered them. A measurement standard cannot depend on an unresolved condition, any more than a standard meter could be defined by an object still being machined. Only a self-contained configuration, identical to itself on every consultation, can function as a ruler mark. The criterion is not chosen to favor State 1; it is what the word *standard* requires. State 7 and State 8 are departures, but they are open — they cannot anchor a scale because they do not hold still. State 1 is the first departure that holds still. The distinction the reader must carry away is exact: **first tilt = State 7 or 8 (open); first resolved mark = State 1 (balanced), reached by three tilts.**

The counting theorem. The three-tilt figure is not an accident of the example; it is a minimum, provable with nothing but this paper's tools. A resolved mark requires a triplet-locked core: three center positions holding one identical active value ([X, X, X] or [Y, Y, Y]). The photon's core is [O, O, O]: zero active values at the centers. Each axis tilt drives exactly one center value from O to an active value — one center move per tilt, by the mechanics of the spin (Paper 3, Section 6). To take the center count of one active value from zero to three therefore requires no fewer than three tilts; two tilts leave a mixed core ([X, X, O] at best), which the identity scan cannot lock. And the conservation constraint rules out any shortcut: the global pool is fixed at three X, three Y, three O, so no tilt can create values — it can only relocate them, one center per move. Hence: **three tilts is the minimum number of center moves that can close a triplet lock while conserving the value pool, and therefore any first resolved configuration must sit at least three steps from the photon.** The first mark's distance from zero is not an observation; it is a counting theorem. The full conserving trajectory is laid out below, with every count shown:

Step	State	Inside (centers)	Outside (corners)	Total	Condition
Start	9 (photon)	O, O, O	3X, 3Y	3X, 3Y, 3O	resolved baseline — identity scan only
Tilt 1	7 (open)	X, O, O	2X, 3Y, 1O	3X, 3Y, 3O	open — lopsided, searching
Tilt 2	5 (open)	X, X, O	1X, 3Y, 2O	3X, 3Y, 3O	open — lopsided, searching
Tilt 3	1 (electron)	X, X, X	3Y, 3O	3X, 3Y, 3O	**resolved — first mark**

Conservation holds at every line — the total never moves from three X, three Y, three O — and the trajectory passes through two genuinely open intermediates before the third tilt closes the triplet lock. The positron trajectory is the exact mirror (through States 8 and 6, driving centers to Y). The full audit with each tilt's mechanics narrated is given in Appendix A; the table above is the result a reader should carry.

3.1 The Symmetry Fallacy and the Triplet Requirement

To anchor the scale of the ruler, the framework must survive a structural stress-test about what actually qualifies as a resolved departure. A superficial reading of the matrix might suggest that any state with an even, symmetrical outer surface qualifies as a stable mark. This raises the problem of State 4, which carries an even outer surface (two X, two Y, two O) even though it sits among the open intermediates. If surface symmetry alone were the test of a baseline mark, the ruler's origin would inherit an immediate tracking error — two candidates for the first mark instead of one.

The framework rejects State 4 as a standalone mark because its external symmetry is an illusion produced by a mixed, unresolved center. State 1 commands a perfect triplet in its center positions — [X, X, X] for the electron or [Y, Y, Y] for the positron. State 4 does not: its core carries one X, one Y, and one O. It has no uniform triplet identity.

Because State 4's center holds one of each value, the core is balanced on paper, and that balance projects a deceptively even outer surface. But the O trapped inside that mixed core is the configuration's identity standard, and without a triplet to lock its orientation it is volatile: it can fluctuate one way toward X or the other toward Y. The state is balanced but has no definitive directional anchor — a condition of structural indecision. It cannot execute a sovereign, non-decaying 720-degree manifold scan, because it has no fixed internal orientation to guide its external temporal sweep. Even outer surface, undecided inner core: symmetry without resolution.

This is the exact contrast that makes State 1 the first mark and disqualifies State 4. State 1's native internal triplet gives it a definitive, unyielding alignment — it knows what it is, and it locks against its outer surface under the Per-Triangle Rule, achieving absolute self-containment on the third axis tilt. State 4 cannot hold its ground as an isolated mark at all.

Two clarifications close off any suspicion that the triplet requirement is invented to fit the electron. First, the requirement is not a new rule added to the framework; it is what the identity scan already demands. The identity scan verifies the center against the surface (Section 2); without a triplet lock, the volatile configuration shifts, causing the scan to return three different answers on successive passes — [X], [Y], [O] by turns — thereby failing to establish a lock. A core of [X, X, X] or [Y, Y, Y] gives the loop a single, unambiguous answer to verify on every pass; a mixed [X, Y, O] core gives it three. The scan can read State 4's core, but it cannot lock it, because there is no one identity there to lock — and a manifold scan whose identity phase cannot lock has no fixed orientation from which to run its temporal phase toward resolution. The disqualification is a consequence of the scan mechanics established in Section 2, before State 4 was ever at issue — State 4 fails by the earlier rules, not by a rule introduced for it. The arithmetic of the failure is two lines: State 1's core is one value taken three times, so every identity pass returns the same answer and the lock closes; State 4's core is three values taken once each, so successive passes return different answers and no lock can close. Even surface does not imply stability when the core fails the identity scan.

Second, State 4 is not pushed off the ledger — it remains a fully functioning open state. It can bond from the outside with the other configurations of the twenty-seven exactly as the other open states do, settling corner against corner where the geometry allows. Nothing special disadvantages it in ordinary bonding. What is special about State 4 is the geometric consequence of its mixed core when three of them converge: the three scattered O values find each other to form a complete [O, O, O] set, the X values and Y values sort into their own triplets, and the merged junction resolves into a photon, an electron, and a positron. That outcome is not a privilege granted to State 4 or a penalty imposed on it; it is the arithmetic of pooling three mixed cores. Three [X, Y, O] centers contain, among them, exactly the makings of one [O, O, O], one [X, X, X], and one [Y, Y, Y] — the convergence simply realizes the triplets the singles could not hold alone (the convergence Paper 3 established, here given its reason).

And the convergence is, necessarily, a multi-configuration event — which connects this section to the conservation argument of Section 4. A lone State 4 cannot resolve itself toward the photon, however close to baseline its even surface sits, because it has no reservoir of free energy with which to execute the transition; resolution is always paid for across an interaction. Its temporal scan therefore searches for the partner or partners that make a conserving resolution possible — whichever the surrounding fabric offers first: an ordinary outer bond with a compatible open state, or, when three State 4 configurations come into proximity, the triple convergence that realizes the triplets. The configuration takes the conserving path the fabric presents. There is no jump to baseline without partners, because there is no energy with which to jump.

So the test for the first mark is not surface symmetry but a resolved, triplet-locked core — the lock the identity scan requires. State 1 passes; State 4, despite its even surface, fails, while remaining a full participant in the bonding ledger. The ruler's origin inherits no tracking error, because only one state besides the photon is self-contained on its own hardware.

4. What "First" Means Exactly

"First" is used in a strict sense, and care is required because two different counts are in play — the count of tilts and the order of resolution. They must not be confused, and the argument of Section 3 turns on keeping them apart.

By the count of tilts, State 1 is not first: it is third, reached after the spin has already passed through State 7 (one tilt) and State 5 (two tilts). The first *tilt* lands on an open state, not on State 1.

By the order of resolution, State 1 is first: it is the first *balanced, self-contained* configuration the spin produces besides the photon. The open intermediates the spin passes through — State 7, then State 5 — are real states, but none of them holds; each is still reaching. State 1 is the first place the spin arrives and stops, resolved.

A natural objection should be answered here directly, because it decides whether the open intermediates are stable enough to be real waypoints at all: if State 7 is open and lopsided, why does it not simply snap back to the photon after the first tilt, rather than completing the sequence? The answer is energy conservation, applied without exception. Every transition between configurations of the twenty-seven-state ledger costs energy — and energy is added geometry, conserved as it relocates, so a configuration cannot spend what was not delivered. The tilt that produced State 7 was driven by creation geometry — energy delivered by the collision event, not generated by the configuration — and once that energy is redirected into the departure, it is conserved there. Reversing to the photon is itself a transition, and a transition requires energy routed through an interaction; a configuration cannot un-tilt itself by fiat any more than it can advance itself by fiat. There is no snap-back because there is nothing to snap back *with*.

One distinction keeps this account fully consistent, and it is worth stating exactly. When the driving event delivers enough energy for the full sequence, the three tilts complete as one continuous driven trajectory: the configuration does not loiter at State 7 mid-collision, because the creation geometry that forced the first tilt is still acting, and it carries the sequence through States 7 and 5 to the resolved State 1 in a single driven pass. The intermediates are then waypoints traversed, not stations occupied. But when a driving event delivers only one tilt's worth of energy — a weaker collision, a glancing interaction — the configuration is *stranded* at the open state the energy reached, and there it holds: lopsided, conserved, searching. The driven sequence and the stranded intermediate are the same mechanics under different energy budgets. Nothing in either case is spontaneous; the trajectory goes exactly as far as the delivered energy carries it, and not one tilt further or back.

This is precisely why the temporal scan exists, and what it is for. An open state cannot resolve alone — resolution is always a multi-configuration event, because the energy accounting of any transition must balance across the participants. The temporal scan is the open configuration's continuous search of the surrounding fabric for the partner, or partners, that make a conserving resolution possible: a matching surplus to settle against, a center to relocate into, a convergence to join. Until the scan returns a partner, the open state holds its departure, lopsided but conserved, sweeping and searching. It does not decay back to baseline on its own, because there is no free energy with which to do so and no interaction in which the books could balance. The framework needs no spontaneous reversals and no unobservable intermediaries to make the ledger close; every step forward in the spin sequence was paid for by the driving collision, and every eventual resolution is paid for by the partner interaction the scan finds. The open intermediates are therefore not fleeting bookkeeping fictions but genuinely held states — open, searching, and stable until a conserving interaction occurs.

It is this second sense — first *resolved* departure — that fixes the ruler. The first balanced state besides the photon is also the smallest balanced departure from zero, so the ruler gains both a *direction* (the order in which resolved departures occur, set by the spin sequence) and a *scale* (the size of the first resolved departure). A ruler with an agreed direction and a fixed first resolved mark can measure. What it must never claim is that this mark is the first tilt; it is the first *resolution*, three tilts out.

4.1 State 7 and State 8: The First Tilts That Set the Axis

Figure 3. The First Tilt Sets the Axis: State 7 (X) and State 8 (Y) Fix the Sign

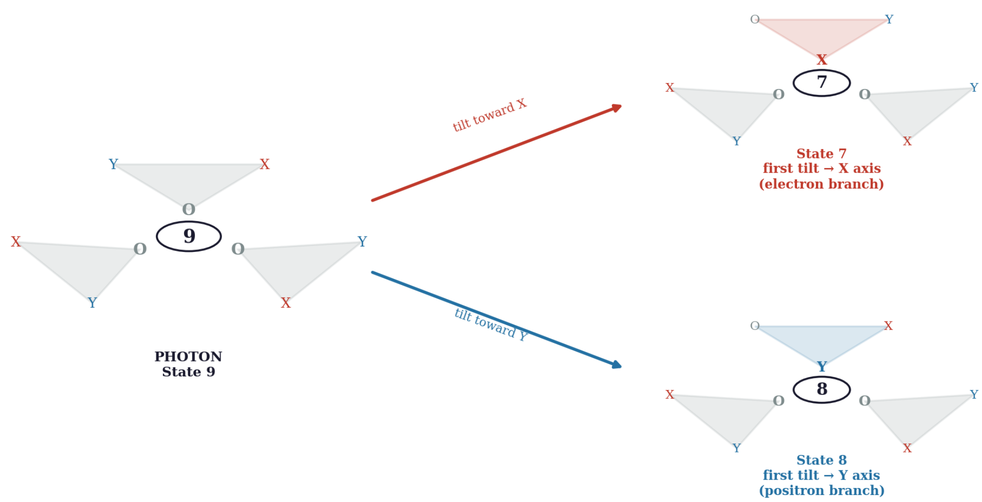

Figure 3. The first tilt off the photon sets the axis: a tilt toward X lands on State 7 (electron branch), a tilt toward Y lands on State 8 (positron branch). The first tilt chooses the sign; the third tilt completes the resolved mark.

Although State 7 and State 8 are not resolved marks, they are not incidental, and they should not be passed over: they are the first single tilts off the photon, and they are what *identify which axis* a departure has taken. This is a structural role worth naming explicitly. The very first tilt off baseline drives one center O toward an active value, and the direction of that first tilt is what commits the whole subsequent trajectory to one sign of the ruler or the other. A first tilt toward X lands on State 7 and commits the sequence to the electron branch (resolving, three tilts out, to [X, X, X]); a first tilt toward Y lands on State 8 and commits it to the positron branch (resolving to [Y, Y, Y]). State 7 and State 8 are the mirror pair that fixes axis identity at the outset.

So the open intermediates carry real structural information even though they are not marks: State 7/State 8 record *which way* the departure went (the sign), and State 5/State 6 are the second tilts continuing each branch toward resolution. The ruler's two signs, named in Section 5 as a property of the resolved mark, are in fact set at the very first tilt — at State 7 or State 8 — and only confirmed when the third tilt completes the resolved State 1. The first tilt chooses the direction; the third tilt makes it a mark.

4.2 The Structural Hierarchy: What the Ruler Measures After the First Mark

Figure 4. The Structural Hierarchy: What the Ruler Measures After the First Mark

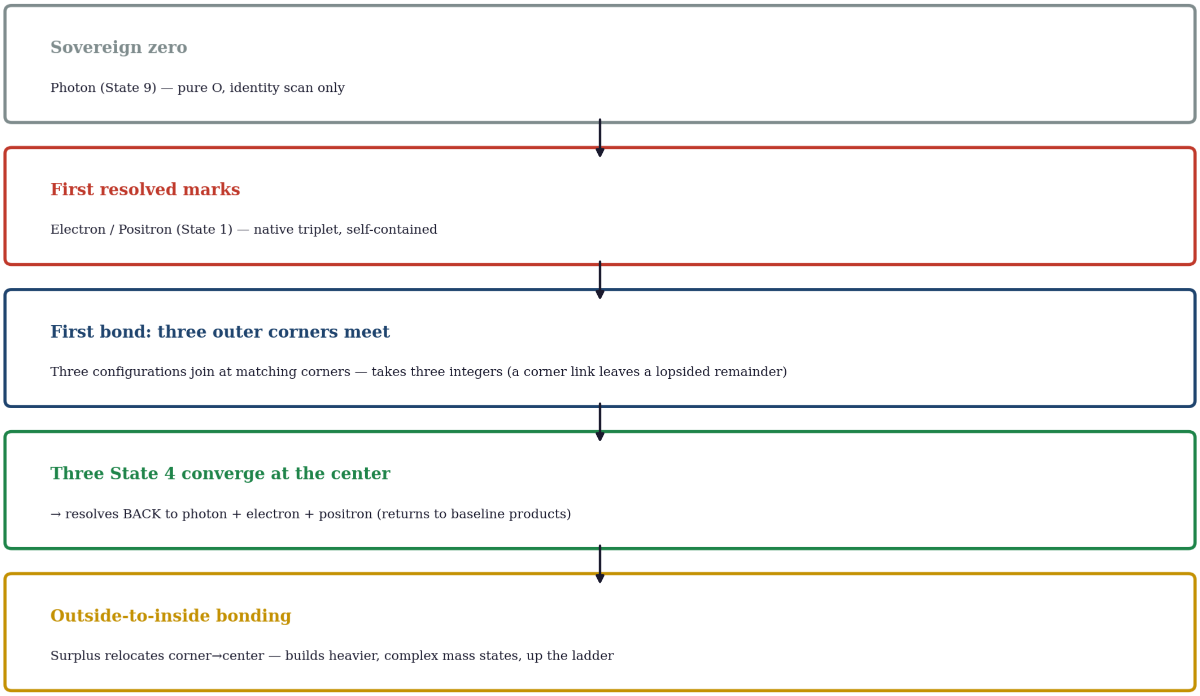

Figure 4. The structural hierarchy: sovereign zero → first resolved marks → first bond (three outer corners meeting) → three-State-4 convergence at the center (resolving back to photon, electron, positron) → outside-to-inside bonding (building heavier complex mass up the ladder).

The first resolved mark is the first unit, but it is not the end of the structural ladder. Laying out the next rungs shows what the ruler is built to measure, in the correct order of increasing structural complexity, without doing the downstream papers' work.

First, the resolved single states: the electron and the positron (State 1), each self-contained on its native triplet. These are the first marks.

Next, the bonding rungs — and here the relationship to Paper 3's published computation must be stated exactly, because the two papers describe two parts of one process, not two competing rules. Paper 3 (Section 8) computed the *first-link pairs*: the value-exchange resolutions in which two open states whose surpluses and deficits exactly mirror (State 5 with State 6, State 2 with State 8, State 7 with State 3) exchange values across their interface and settle both outer surfaces toward the balanced baseline. That computation stands, and it is the first part of the bonding process: the exchange that the matrix permits and forbids. What this paper adds is the mechanics of the *physical joint* itself. A value exchange settles the ledgers, but a lasting composite requires the configurations to join at a shared outer junction — and a corner-to-corner joint between two alone does not close cleanly: the shared junction carries a lopsided remainder. The exchange is therefore only the opening of the bond. Either further participants arrive to balance the shared junction — three configurations meeting at their matching outer corners is the first arrangement that settles the joint completely — or the participants part and the link does not hold. Paper 3 computed which exchanges are permitted; this paper describes the joint mechanics in the detail the ladder requires; and the full composite spectrum that these joints assemble into is developed in Paper 16. The first completed bonding rung is therefore the three-corner junction — three individual integers settling together at their corners after the permitted exchanges open the link.

Next, and structurally special, the three-State-4 convergence at the center. Three State 4 configurations meet not at their corners but at a shared central junction, pooling their mixed cores into complete triplets (Section 3.1), and resolve into a photon, an electron, and a positron. This one runs the other way on the ladder: rather than building upward into something heavier, it resolves *back down* to baseline-level products — the photon returning to the sovereign zero, the electron and positron to first marks. It is the rung where complexity converts back into the baseline and the first marks, not into heavier mass. And it is no violation of the no-snap-back rule of Section 4, precisely because it is a three-body interaction: the resolution is paid for across the interaction, the energy accounting balancing among the three participants — which is exactly what a lone configuration, with nothing delivered and no partner to balance against, can never do.

Next, outside-to-inside bonding — the rung that builds upward. When a surplus value relocates from an outer corner into a center, the bond drives a value from the outside to the inside, loading the core further. This is the mechanism by which heavier, more complex mass states are assembled: each outside-to-inside relocation deepens the inside, climbing the ladder toward composite matter. Where the three-corner bond settles surfaces and the triple convergence resolves back to baseline, outside-to-inside bonding is what accumulates mass upward.

This is the structural hierarchy the ruler measures along: sovereign zero (photon), first resolved marks (electron/positron), first bond (three outer corners meeting), three-State-4 convergence at the center (resolving back to photon, electron, positron), and outside-to-inside bonding (building heavier complex mass up the ladder). Paper 4 fixes the zero and the first mark; the bonding and composite rungs are computed in Paper 3's matrix and developed for the full spectrum in Paper 16. Naming the ladder here shows that the ruler is not a single mark but a graded scale, each rung a distinct resolved operation.

5. Mass and Direction as the First Two Ruler Readings

With zero and first resolved mark fixed, the first two things the ruler reads follow, and naming them sets up the papers that develop them.

Mass is the first reading. Paper 3 established that departure registered on the inside *is* mass; the photon's pure-O inside is the zero of mass, and the loaded inside of the first resolved state is the first nonzero mass reading on the ruler. The full accounting of mass as departure from c is the subject of Paper 8, and this paper does not pre-empt it — it marks only that the first resolved mark is where the mass scale begins.

Direction is the second reading. The spin runs two ways from baseline: one sequence resolves to the electron, the mirror sequence to the positron. These are the ruler's two signs — not two substances, but one resolved departure reached in two rotational directions (consistent with Paper 2.5's position that antimatter is tilt direction, not separate substance). The ruler thus reads not only *how far* a configuration sits from baseline but *which way* it departed.

6. One Medium, Not Many Fields

A consequence worth stating plainly, because it is constructive rather than critical, is that this baseline removes the need for separate fields stacked on top of space. In the framework there is one medium — the three-axis fabric — and one kind of departure from its baseline: the axis spin. What standard physics distributes across a separate electromagnetic field with its own carrier is, here, the photon itself and the tilting of the single medium. The photon is not an excitation of a field laid over space; it is the baseline state of the fabric, and the electron and positron are that same fabric tilted into a resolved departure. One medium, tilted, accounts for what the standard picture needs a dedicated field to carry.

This paper makes only the positive claim: one baseline and one medium suffice, so no separate field is required. The demonstration that the separate-field machinery is not merely unnecessary but actively generates the shifting-scale problems catalogued below is the work of the companion critique, Paper 4.5.

7. Structural Scale Invariance: The Per-Triangle Rule as Absolute Baseline

The final property that makes this a *ruler* and not just a model is that its scale cannot be adjusted. The Per-Triangle Rule — every triangle carries exactly one X, one Y, one O — fixes the geometry of the fabric from first principles, with no free resolution parameter. There is no grid spacing to choose, no scale to tune, because the structure is not a numerical scheme laid over space; it is the fabric's own hardware. The discrete nature of the fabric's interlocking triangles does not constitute an arbitrary grid spacing, because the geometry is a fixed, non-tunable boundary condition of the hardware itself rather than a computational parameter chosen to approximate a limit. Every three-triangle junction must balance its X, Y, and O values against its immediate neighbors, with no loose variables, no remainder, and nothing requiring removal by hand.

When the photon resolves into the electron-positron state through the spin, the resulting balanced configuration is a fixed, physical boundary of the medium — the first resolved mark — whose size is dictated by the hardware of the fabric, not by any choice of resolution. It is therefore scale-invariant in the strict sense: the same regardless of how finely or coarsely anything external is reckoned, because nothing external sets it. This is what an absolute baseline buys that a tunable scheme cannot: a first mark that no programmer, and no fit to data, can move.

This paper states the property positively. The contrast with the standard computational approach — which lacks an absolute baseline and must therefore choose a scale, tune it by hand to a known value, and extrapolate toward a limit — is drawn in full in Paper 4.5.

7.1 The Failure of the Lattice Fallacy: Creation Geometry and Dynamic Expansion

To evaluate this framework through the lens of traditional lattice QCD or discretized computational models is to commit a fundamental category error. A reviewer accustomed to standard numerical physics may look at the discrete, interlocking triangles of the fabric and immediately attempt to define a variable grid spacing parameter (delta-x) to tune the resolution of the mathematics. This approach is profoundly incompatible with the mechanics of a continuous, self-contained medium for three structural reasons.

1. The Exponential Explosion of Creation Geometry. In standard lattice simulations, a fixed number of fields or particles are modeled interacting within a passive, empty coordinate grid. In this framework, space is not an empty container; the fabric is the matter.

Every single time an interaction occurs within the medium — whether it is a high-speed collision or a local geometric resolution — there is an inherent probability of triggering creation geometry. An interaction does not merely shift values between static points; it actively tilts axes, relocates structural currency between internal cores and external surfaces, and shifts configurations among 27 distinct geometric states.

If a programmer attempts to model this framework using a traditional tunable grid, they cannot simply adjust a single spacing parameter to smooth out the infinities. Because every collision dynamically alters the local hardware layout itself, the mathematical ledger would be forced to compute an exponential explosion of shifting, interdependent grid definitions simultaneously. The math instantly deconstructs and blows up the ledger.

2. Temporal Scan Driven Expansion. Furthermore, a static grid cannot account for the dynamic behavior of duration within the medium. When a configuration departs from the baseline photon state, the redirected energy forces it to execute an outer temporal manifold scan.

This temporal scan is the mechanical mechanism of expansion. The fabric actively stretches its processing cycles outward to audit and reconcile its localized displacement against the surrounding medium. To attempt to fit this fluid, expanding processing loop into a rigid, pre-programmed lattice spacing is impossible. One cannot use a fixed mathematical cheat-code to measure a boundary that is actively defining its own geometric footprint in real-time.

3. Absolute Hardware vs. Tunable Cheats. A tunable grid is an artificial, mathematical tool chosen by a programmer to approximate smooth physics and hide infinities by taking a limit (delta-x to zero). In contrast, the discrete triangles of this framework are absolute, non-tunable physical hardware.

The Per-Triangle Rule (1X, 1Y, 1O) is an unyielding geometric boundary condition of reality, not a variable slider on a simulation dashboard. There are no free parameters to twist, no scaling factors to adjust, and no hidden infinities to normalize by hand. There is no option to "manipulate the answer" to fit experimental data. The system is entirely self-contained, scale-invariant, and deterministic. It is what it is.

The three reasons compound into a single verdict. The lattice reflex — see something discrete, reach for a spacing parameter — mistakes hardware for software. A simulation's grid exists to be adjusted, because it is an approximation scheme standing in for a continuum the equations cannot otherwise handle; the moment it stops being adjustable it stops being useful to the simulator. The fabric's triangles are the opposite kind of thing: they are not standing in for anything, they approximate nothing, and their fixedness is not a limitation but the very property that makes the universal ruler possible. A reviewer who asks for this framework's delta-x is asking which slider controls reality's resolution — and the framework's answer is that there is no slider, there is no dashboard, and that is precisely the point. The detailed audit of what the tunable-grid method must do to reach physical answers — impose, decree, tune, extrapolate — is the work of the companion critique, Paper 4.5, Section 4.

8. The Ruler Set, Pointing Forward

Paper 4's work is to plant the anchor, not to do the measuring the rest of the series is built for. With the origin and scale fixed, the downstream papers each measure a different quantity against this same baseline:

- Configuration stability — why some departures hold and some must resolve — is Paper 5.
- Rest, uniform motion, and inertia as internal ledger states — Paper 6.
- Mass as the internal accounting of departure from c — Paper 8.
- Gravity as resolution-seeking and departure from the photon baseline — Paper 10.
- Time as the 720-degree rotation ledger — Paper 13.

Each reads its quantity from the zero this paper fixed, in units scaled to the first resolved mark this paper identified. Paper 4 is the anchor they all measure from.

9. Conclusion

The photon is the zero of the universal ruler: no departure, no mass, no duration — it runs only the inner identity scan, and duration is the accumulation of temporal scans it never runs. The electron-positron pair is the first resolved mark: not the first tilt — a single tilt lands on the open State 7 or State 8 — but the first balanced, self-contained configuration besides the photon, reached after three tilts through conserving intermediate states, and taken in two rotational directions that fix the ruler's two signs. The distance is a theorem, not an observation: three tilts is the minimum number of center moves that can close a triplet lock while conserving the value pool, so any first resolved configuration must sit at least three steps from the photon. The baseline is absolute because the Per-Triangle Rule fixes the fabric's geometry with no tunable scale, so the first mark is a physical boundary of the medium rather than a chosen parameter. With an origin and a scale now set, the ruler Paper 1 promised has the two things it needed before it could measure anything. The papers that follow do the measuring; this one fixed what they measure from — and froze the vocabulary they measure with (Appendix B).

References

This paper is a Tier 1 foundational paper of the *Departure From the Photon Baseline* series and builds on the framework established in the companion papers below.

1. Kadunic, D. (2026). *The 27 Lagrangians of Gravity: A Survey of the Field's Standards of Evaluation* (Paper 1). Zenodo. https://doi.org/10.5281/zenodo.20434286
2. Kadunic, D. (2026). *The Three-Axis Framework and the Nine-State Classification of the 27 Geometric States* (Paper 2). Zenodo. https://doi.org/10.5281/zenodo.20510019
3. Kadunic, D. (2026). *They Assumed It; They Did Not See It: A Critique of Inferred Particles in Standard Physics* (Paper 2.5). Zenodo. https://doi.org/10.5281/zenodo.20518514
4. Kadunic, D. (2026). *The Nine-State Classification with Inside-Outside Structure* (Paper 3). Zenodo. https://doi.org/10.5281/zenodo.20561941

Paper 1 establishes the photon baseline as the universal reference and the need for a single ruler. Paper 2 establishes the three-axis framework, the nine-state classification, the radiation-trefoil configuration, and the Per-Triangle Rule. Paper 2.5 establishes the framework's position that antimatter is tilt direction rather than separate substance (Section 5). Paper 3 establishes the inside-outside structure, the spin operation, the conserving spin trajectory (Appendix C), and the bonding and triple-convergence mechanics (Section 8) on which the structural hierarchy of Section 4.2 draws. The mass accounting named in Section 5 is developed in Paper 8; the time treatment of Section 2.2 is developed in Paper 13; the bonding and composite rungs of Section 4.2 are developed in Paper 16. The critique of the standard model's in-flight attribution of photon properties (Section 2.1) is the work of the companion critique, Paper 4.5.

A. The Comprehensive First-Resolved-Mark Audit Ledger

This appendix walks the spin from the photon baseline to the electron as a strict conservation audit, one tilt at a time, so the arrival at the first resolved mark can be verified without exception. The discipline of the audit is the point: where the standard calculation of an electron's properties requires summing an open-ended series of corrections (the five-loop QED expansion alone runs to 12,672 diagrams, as Paper 3.5 documents), the framework tracks the entire departure with a single three-axis conservation ledger that never produces a deficit or a remainder. Every line below conserves three X, three Y, three O across the nine positions.

The photon baseline is inside [O, O, O], outside [three X, three Y]. The target electron is inside [X, X, X], outside [three Y, three O]. The spin reaches it through three discrete axis tilts; at each tilt one triangle's center value is driven from O to X, and by the Per-Triangle Rule that triangle's two outer corners re-settle to the remaining two values.

- **Start — State 9 (photon).** Inside [O, O, O]; outside [three X, three Y]. Inside count: 3 O. Outside count: 3 X, 3 Y. Total: 3 X, 3 Y, 3 O. The configuration is balanced and runs the identity scan only; no temporal scan, no mass, no time.
- **After tilt 1 — State 7 (open).** One center O is driven to X. Inside [X, O, O]; the tilted triangle's corners re-settle so the outside becomes [two X, three Y, one O]. Inside count: 1 X, 2 O. Outside count: 2 X, 3 Y, 1 O. Total: 3 X, 3 Y, 3 O — conserved. This is the **first tilt**, and it lands on an **open** state: the outer surface is now lopsided (a surplus and a deficit are exposed), so the configuration reaches for resolution rather than holding. It is a departure, but not a mark.
- **After tilt 2 — State 5 (open).** A second center O is driven to X. Inside [X, X, O]; outside re-settles to [one X, three Y, two O]. Inside count: 2 X, 1 O. Outside count: 1 X, 3 Y, 2 O. Total: 3 X, 3 Y, 3 O — conserved. Still open, still lopsided, still reaching. Two tilts out, still no resolved mark.
- **After tilt 3 — State 1 (electron, resolved).** The third center O is driven to X. Inside [X, X, X]; outside re-settles to [three Y, three O]. Inside count: 3 X. Outside count: 3 Y, 3 O. Total: 3 X, 3 Y, 3 O — conserved. The outer surface is now cleanly split with nothing unsettled: the configuration is **balanced and self-contained**. This is the first resolved mark — reached on the third tilt, not the first.

Three facts close the audit. First, conservation is exact at every line: the total is three X, three Y, three O throughout, so nothing is created or destroyed; values only relocate between inside and outside as each triangle re-settles. Second, the route to the first resolved mark runs through two genuinely open states (State 7, then State 5) before arriving at the balanced State 1 — the mechanical demonstration that State 1 is the first *resolved* departure and not the first tilt. Third, the open intermediates hold rather than reverting: a configuration at State 7 cannot snap back to the photon, because reversal is itself a transition and every transition costs energy the configuration does not privately possess. The tilts of this audit were paid for by the driving collision (the creation geometry of Section 6, Paper 3); the conserved energy now sits in the departure, and the only path anywhere — forward or back — runs through a conserving interaction with a partner that the temporal scan must first find (Section 4). The positron audit is the exact mirror, driving centers from O to Y and passing through State 8 then State 6 before resolving to State 1 (positron). No step requires a free parameter, a chosen scale, a removed infinity, or a spontaneous reversal; the single three-axis ledger closes on itself.

B. Frozen Glossary of Series Terms

The terms below are the fixed currency of this series. They are defined here once, with the paper of origin noted, and these definitions do not shift in later papers. Subsequent papers reference this glossary rather than redefining.

Configuration. Three triangles meeting at a shared central junction, occupying nine positions: three center (the inside) and six outer-corner (the outside). (Paper 2.)

Per-Triangle Rule. Every triangle carries exactly one X, one Y, and one O; a triangle's center value forces its two outer corners to the other two values. The single axiom of the geometry. (Paper 2; motivated in Paper 3, Section 2.)

The inside / the outside. The three center values (identity, mass) and the six outer-corner values (interaction surface), respectively. The inside fixes the outside's content counts. (Paper 3.)

Resolved. A configuration whose outer surface carries nothing unsettled — balanced or cleanly split — and whose core the identity scan can lock; self-contained, requiring no partner. States 9 and 1 only. (Papers 2, 3.)

Open. A configuration carrying an uneven outer surface and an unlockable or displaced core; it cannot stand alone and searches for resolution. States 2 through 8. (Papers 2, 3.)

Identity scan. The inner 360-degree loop by which a configuration verifies its three center values against its six outer values — confirming what it is. The photon runs this scan alone. (Paper 2, Section 1; Paper 3, Section 7A; this paper, Section 2.)

Temporal scan. The outer 360-degree loop, present only after departure, by which a configuration audits the surrounding fabric for the partner or partners that make a conserving resolution possible. (Paper 2, Section 1; Paper 3, Section 7B; this paper, Sections 2 and 4.)

Manifold scan. The complete 720-degree cycle: identity scan plus temporal scan, run by every configuration that carries mass. (Paper 2, Sections 1 and 4.1, Appendix B; this paper, Section 2.2.)

Duration (time). Operationally defined as the accumulation of completed temporal scans. A configuration that runs no temporal scans accrues no duration. (This paper, Section 2.1; full ledger treatment in Paper 13.)

Mass. The departure from the photon baseline registered on the inside — the ledger entry tracking how much energy has been transferred into the localized displacement of the center positions. (Papers 1–3; this paper, Section 2.2; full accounting in Paper 8.)

Energy. Added geometry: the structural content delivered into the medium by an interaction, conserved absolutely as it relocates. A configuration cannot spend what was not delivered. (Series foundations; this paper, Section 2.2.)

Creation geometry. The localized geometric layout instantiated by a collision or structural event, which forces axis tilts as its deterministic consequence. The driver of all transitions; nothing tilts by fiat. (Paper 3, Section 6.)

Axis tilt / spin. The operation that drives one center value to another value, relocating values between inside and outside under the Per-Triangle Rule while conserving the global pool. (Paper 3, Section 6.)

Sovereign zero. The photon (State 9): pure O inside, balanced outer surface, identity scan only — zero departure, zero mass, zero duration. The ruler's origin. (This paper, Section 2.)

First resolved mark. State 1 (electron or positron): the first balanced, self-contained configuration besides the photon, reached by a minimum of three tilts (the counting theorem, Section 3). The ruler's first unit. (This paper, Section 3.)

Triplet lock. The condition of a core holding one identical active value at all three centers ([X, X, X] or [Y, Y, Y]), giving the identity scan a single answer to verify on every pass. Required for resolution; impossible for a mixed core. (This paper, Section 3.1.)

Driven sequence / stranded intermediate. A trajectory whose tilts complete in one continuous pass under sufficient delivered energy, versus a configuration held at an open state when the delivered energy reached only part way. Same mechanics, different energy budgets. (This paper, Section 4.)